

Beware, win or lose

Domestic violence and the World Cup

A six-pack, the big match on television – and in the wake of alcohol-fuelled emotion, violence against a partner. Did domestic violence increase when the World Cup was on? **Allan Brimicombe** and **Rebecca Cafe** tested the truth of an official analysis.

The Football World Cup. The Champions League. The Rugby World Cup. Super Bowl Sunday. In the run-up to just about any big sporting event these days there are warnings and public conversations about a likely rise in domestic violence while the sport is on. London 2012 was no exception: Figure 1 shows a beer mat used in the London Borough of Newham (one of the London 2012 “host boroughs”) as part of its campaign in the run-up to and during the Olympic Games.

The World Cup is of course the biggest event in football, or soccer if you prefer. Many of these conversations, in the UK at least, quote from news reports of a Home Office analysis¹ which purports to have found a link during the 2006 World Cup between sporting events, alcohol consumption and a rise in domestic violence. “Major sporting events do not cause domestic violence, as perpetrators are responsible for their actions”, said the analysis, “but the levels of alcohol consumption linked to the highly charged emotional nature of those events seems to increase the prevalence of such incidents.”

Was the Home Office report justified? Does domestic violence really go up when big sports matches are on?

Campaigners would argue that domestic violence is a serious everyday event not caused by sport or alcohol and that public consciousness of this scourge should not just be tied to particular sporting fixtures or tournaments. Spikes in it can be much larger, for example over Christmas and New Year, than at times of big matches. Nevertheless, the perennial nature of the debate, and possible new data from the 2010 World Cup, prompted one author of this paper, a BBC journalist, to put the 2006 Home Office conclusions to the test² (though the BBC did not run the story until the time of Champions League earlier this year). She invited the other author, a statistician, to help with the analysis.

For reasons we shall discuss below, police-recorded figures of domestic violence are more problematic than, say, burglary or vehicle theft.

Domestic violence is defined under the Domestic Violence Crime and Victims Act, 2004 as “any incident of threatening behaviour, violence or abuse (psychological, physical, sexual, financial or emotional) between adults who are or have been intimate partners or family members, regardless of gender or sexuality”. “Adults” are defined as persons aged over 18 and “family members” include mother, father, son, daughter, brother, sister and grandparents, whether directly related, in-laws or step-family. Domestic violence extends to female genital mutilation, forced marriage and so-called “honour

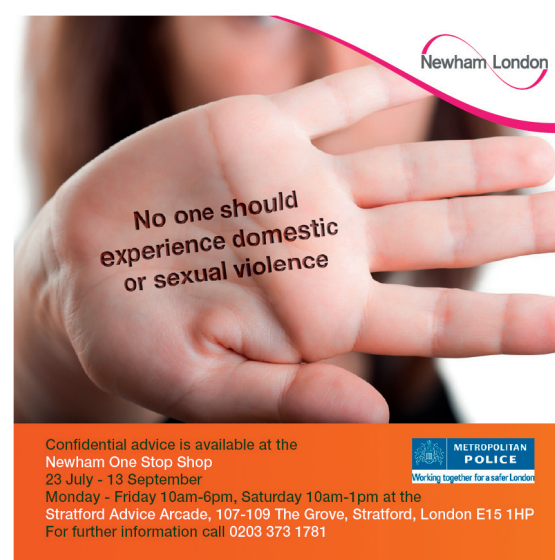


Figure 1. Beer mat design used in an information campaign about domestic violence during the 2012 Olympic Games (courtesy of Safer Newham Partnerships, London Borough of Newham)

30% of women and 17% of men have experienced domestic abuse. Does the rate go up when big sports matches are played?

crimes". Other general terms in use for it are domestic abuse, inter-partner violence and intimate violence.

Since from its definition it can be a wide range of different behaviours, there is no specific statutory offence of domestic violence. Nevertheless, many forms of domestic violence are crimes and where reported to the police will be recorded under a number of categories such as assault, criminal damage, rape or murder, depending on the outcome of the incident and Home Office rules for recording crime. Furthermore, because domestic violence is not a statutory offence, it is not reported by police forces to the Home Office and does not appear separately as a category in the published crime statistics. In this respect domestic violence is "below the radar" of national statistics. However, police forces do ensure that where a case is recorded as a notifiable crime, a flag or some other database entry is made to signal it as having been domestic violence.

Database designs for recorded crime vary across police forces. Furthermore, notifiable crimes are only about 20% of all incidents recorded by the police³, and where the circumstance of a call-out for domestic violence is not deemed to constitute an offence it may be recorded as a domestic incident (such as heated argument, or abuse).

The upshot of all this is that although it is not always straightforward, statistics on domestic violence can in fact be retrieved from police databases. But they are not published, and, as we shall see, it takes a request – in fact, many requests – under the Freedom of Information Act to get them.

The Crime Survey for England and Wales has included since 2004/05 a self-completion module on domestic abuse and intimate violence which is the only reliable measure easily available of national prevalence and trends. The 2010/11 survey⁴ showed that 7% of women and 5% of men had experienced domestic abuse over the previous 12 months, equivalent to 1.2 million female victims and 0.8 million male victims. (This is a 60–40 split. Readers may be surprised, as we were, at the ratio. The common perception is that violence in the home is overwhelmingly directed against women.) Overall, 30% of women and 17% of men had experienced domestic abuse since the age of 16. The 2010/11 figures are the lowest since 2004/05 and show no statistically significant change since 2008/09. This would suggest an overall decline in domestic violence that has

in recent years flattened out. Nevertheless, the Home Office estimates¹ that domestic violence accounts for 15% of all violent crime and for 35% of murders. For women, the risks of physical assault were highest for those aged 16–24, separated from their spouse, for Council tenants, and for those in poor health and/or in financial difficulties. For men, the risks were highest for those aged 16–24, cohabiters, the unemployed and, again, those in financial difficulties⁵. Victims were found to have far higher levels of alcohol consumption than non-victims, but that may be either a cause or a consequence of victimisation.

The overall proposition we were testing was whether or not football tournaments are associated with an increase in the risk of domestic violence. It is not that football tournaments *cause* the violence, but rather that

The excitement and adrenalin from watching a national team play may exacerbate tensions within a relationship, leading to loss of temper and to violence

the excitement, disappointment and flow of adrenalin resulting from watching a national team play may exacerbate existing tensions within a relationship and result in lost tempers and violence or abuse. Such behaviour may be made worse or more likely when alcohol has been consumed.

The Home Office study of the 2006 World Cup¹ used data from 46 basic command units from 17 police forces in England and Wales (there are a total of 219 command units across 43 forces). The study took both incident and offences data on the six match days when England played, and compared them with a control period from February 3rd to March 31st of that same year during which an enforcement campaign against domestic violence was being carried out. There are problems with this. The control period was in winter; the World Cup was played in summer. Seasonal effects might have made the comparison invalid. Furthermore, the enforcement campaign might have biased the police forces in their recording of domestic violence. The command units chosen were also mostly in urban areas. The

Home Office findings were that while the per-match increase in *incidents* ranged from +4.6% to +31.4%, the per-match change in *offences* ranged from –11.3% to +28.9%.

Importantly, the study did not differentiate between the *outcome* of each match – win, lose or draw for England. It seems intuitive that a win would fill an England fan with cheerfulness and goodwill towards the world in general, and a loss would have the reverse effect. Our own study found that the result of the match was indeed an important factor – though not quite in the way that we might have expected.

The Home Office can obtain data from police forces through internal channels. Ordinary citizens like ourselves must request such data under the terms of the Freedom of Information Act 2000. Consequently, for our story, the BBC made a Freedom of Information (FoI) request for data on domestic violence between the dates of 11 June and 11 July 2010 (which was the period of the 2010 World Cup). A separate FoI request had to be made to each of the 52 police forces in the United Kingdom. This was repeated for the 2006 World Cup dates. Later, when the statistician author came on board, he identified that a stronger control period was needed and further FoI requests were made for equivalent dates in 2005 and 2009, that is, the year preceding each World Cup. Thus a total of 208 FoI requests had to be made in order to carry out this study – a time-consuming effort which others contemplating FoI requests of a similar nature should note.

We needed to find a suitable control period against which to compare any change. This was particularly problematic. As we have hinted, comparing the amount of violence at times of football matches to the amount at Christmas would give the misleading impression that violence goes down at big-match times. We have to compare like with like.

To understand some of the issues around finding a suitable control some historic data are graphed in Figures 2 and 3. Figure 2(a) shows the daily count of domestic violence offences for London for the financial year 2000/01 against the daily average of 131 offences. There is a strong short-term periodicity which is a weekly cycle. (Weekends are bad times for domestic violence – see below.) The largest spike is at New Year, with a build-up over Christmas. So much for the season of goodwill. Figure 2(b) shows a 14-day moving average of 16 months of data to illustrate other patterns,

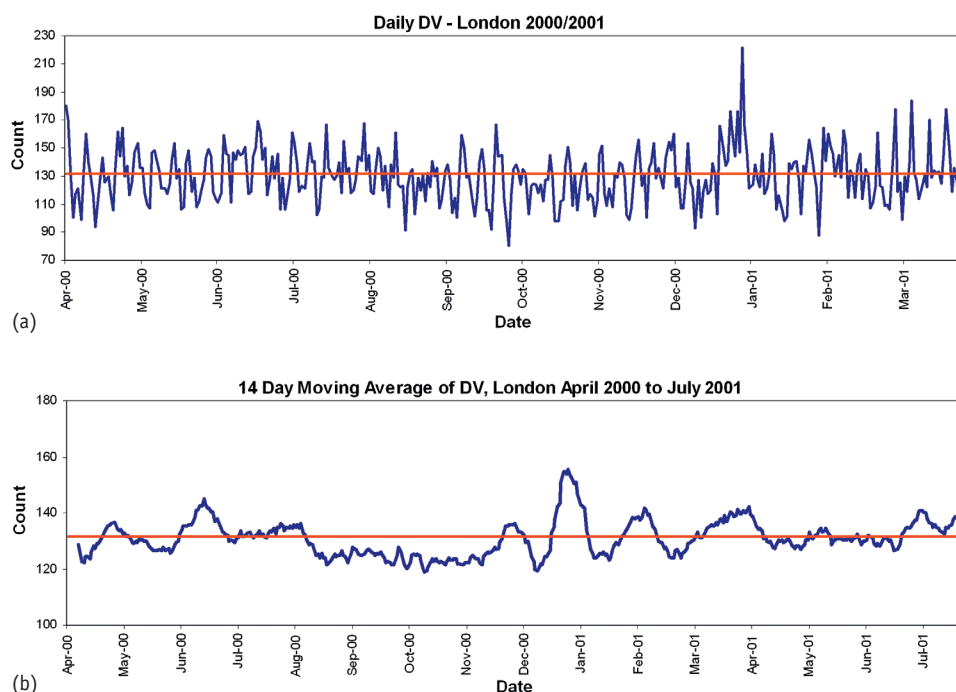


Figure 2. Domestic violence data for London: (a) daily count of domestic violence against the daily average for 2000/01; (b) 14-day moving average for 16 months from April 2000

not all of which are easily explained. A peak in June 2000 corresponds with the period of the 2000 European Cup; there was no peak in June 2001, a non-Cup year, and summer violence was close to or even below the average for the period. Figure 3(a) gives the percentage of domestic violence that is reported on each weekday. It is higher at the weekend and lower during mid-week. This is reinforced in Figure 3(b) which shows the percentage change on the previous day: there is a sharp increase on Friday, with peak increase on Saturday. Reporting of violence may not correspond to the day on which it happened, and it may well be that the relatively high Monday figures include residual reporting from the weekend. Clearly, then, any control period for domestic violence

must be for the same days of the week and should preferably be for the same time of year.

Given that the data for this study was based on 208 separate FoI requests, it will come as no surprise that there were data consistency problems. There were no returns at all from Lincolnshire, some forces had changed their databases since 2006 and could not return the earlier data, and for some the totals for each period seemed such a contrast (sometimes by an order of magnitude) as to be considered dubious. Returns for the City of London (that is, the financial district, which has a very small residential population) were in single digits and would produce small-data problems in the analysis, and might result in the identification of individuals if included. Our intention had

been to rerun the Home Office analysis of the 2006 World Cup using data from many more Forces; but these missing and inconsistent data would have made our results insufficiently robust. So instead we focused on the effect of England matches in the 2010 World Cup. We used data from 33 out of 39 police forces in England, which covered 77% of England's population. The dates of matches played by England and the results are given in Table 1. (American soccer fans may well remember their draw against England in the first round. That particular World Cup, it has to be said, was not a particularly glorious one for England.) The domestic violence reported by the police forces on those days is also given, as a count and in rates per thousand adults.

Whilst there is a clear contrast in the national total of domestic violence between the Friday draw and the Sunday loss, there seems little to separate the Saturday draw and the Wednesday win. But as Figure 3 showed, Wednesdays normally have less violence than Saturdays, so for the Wednesday total to exceed a Saturday one implies that it is elevated. Rates from different police forces for the Saturday showed that rural Cumbria (21 incidents, 0.051 per thousand population) and metropolitan London (375 incidents, 0.060 per thousand) are not so dissimilar, and no urban–rural difference was evident, so we have not listed the figures for each of the police forces separately.

The analysis was to be the basis of a BBC news article for the general public. Its results therefore had to be simple enough for lay people to understand, while the article itself had to be robust enough to stand up to scrutiny by expert third parties.

The original thinking had been to test the 2010 World Cup against the equivalent days in 2009. The 2010 event started on Friday, June 11th and ran for 31 days to the final on Sunday, July 11th. The equivalent days in 2009 are from Friday, June 12th to Sunday, July 12th. This ensures that each day of the week appears the same number of times in each period and individual days can be compared. However, the comparison of June 2000 and June 2001 in Figure 2 suggests that the effect, if it exists, is broader than day-by-day and spreads out beyond just the individual match days. Violence may not be reported to the police on the day it occurs, or may not occur on the match day but, say, the day before or after, so its incidence over the tournament period will be diffuse. So we ran a check using a more stringent control. We

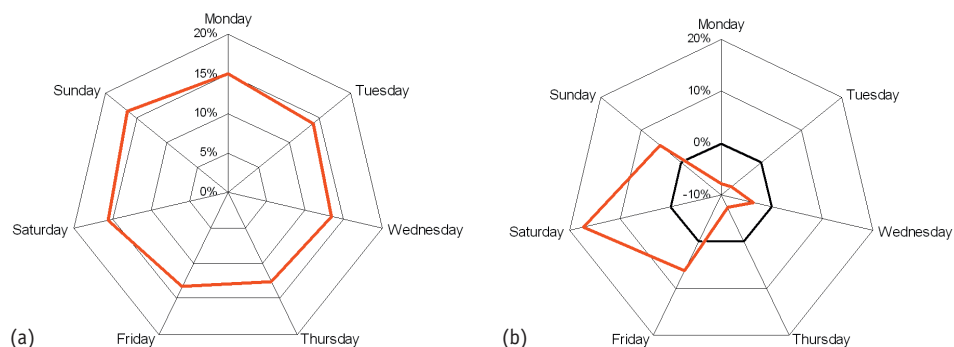


Figure 3. Domestic violence by weekday for London 2000/01: (a) each day as a percentage of all domestic violence; (b) each day as percentage change on previous day

Table 1. England fixtures during the 2010 FIFA World Cup, and domestic violence count and rates on match days. Rates are per thousand population aged 16 and over

Match date	Saturday, June 12th	Friday, June 18th	Wednesday, June 23rd	Sunday, June 27th
Fixture	England v. USA	England v. Algeria	England v. Slovenia	England v. Germany
Result	Draw (1–1)	Draw (0–0)	Win (1–0)	Lose (1–4)
Domestic violence count (national total) and rate (per thousand)	2539, 0.065	1980, 0.051	2557, 0.065	3469, 0.088

compared the Saturday 2010 match day with the other four Saturdays of the 2010 Cup – Saturdays on which England was *not* playing – and we took their averages, recorded by 33 police forces, as our stronger control. We did the same for the other England match days.

Table 2 summarises our results. We know, from the Crime Survey for England and Wales⁴, that domestic violence in 2010 was at its lowest since 2004; yet the non-match days of our 2010 control period showed a 10–16% increase in domestic violence over 2009. (This is a significant difference at the 95% confidence interval, except for the Sunday). This most probably reflects the diffuse effect we mentioned above – the generalised increase, spread out over several days, of a summer with a football tournament as opposed to one without.

We turn next to days on which matches were held. On days when England drew there was no significant increase in the rate of reported domestic violence. On days when England either won or lost there was a significant increase. These results held whether we compared 2010 match days to 2010 non-match

days, or to the 2009 non-tournament year. Whichever of our two control periods we used, we had to come to the same conclusion.

The results would seem pretty irrefutable. They may also come as a surprise. It is not unexpected that everything depends on the outcome of the match, but the way that dependency shows itself is not so expected. Violence in the home on big match days certainly increases if England lose. It certainly also goes up on big match days if England win. Win or lose, there will be a significant increase in the rate of reported domestic violence. But if England draw, the rate of domestic violence does not increase. One might have thought that a draw would heighten frustration, which might be taken out on partners; psychologists would have to tell us if instead it flattens emotion to a more apathetic state.

The Home Office analysis was therefore only partly right. Domestic violence is associated with national football tournaments, but only if there is a definite win or lose result. Its failing was that it ignored the outcome of the match, which, as we have seen, is crucial.

The percentages we found are so great that we would be able to rest our case. When the BBC piece² was aired the result was a flurry of emails that both congratulated and challenged the findings; no doubt the debate will rumble on. Of course we realise that we have no evidence at an individual level that all or any of the reported domestic violence cases in 2010 resulted from or could be associated with watching England fixtures. These are aggregate police force level figures. Nevertheless, if it was not the football, then what was it? We can think of no other event occurring on the Wednesday of England's win and the Sunday of England's exit from the World Cup that would explain these significant increases in the rate of reported domestic violence.

References

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Should any reader feel affected by the content of this paper, the UK National Domestic Violence Helpline is 0808 200 0247; the US National Domestic Violence Hotline is (800) 799 7233. Anyone in immediate danger should always phone the emergency services.

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Table 2. England in the 2010 World Cup: results of significance testing on proportional change in rate of domestic violence between A: 2009 and 2010 non-match-day control; B: England match days and 2009 control; C: England match days and 2010 non-match-day control

Day	Saturday	Friday	Wednesday	Sunday
Result, if a match day	Draw	Draw	Win	Lose
<i>Difference between 2009 control and 2010 non-match-day control</i>				
A: Proportional change in rate	15.9% $t = 2.16, 29 \text{ df}$ $p = 0.039$	13.5% $t = 2.14, 29 \text{ df}$ $p = 0.041$	14.6% $t = 2.36, 29 \text{ df}$ $p = 0.025$	10.1% $t = 0.88, 29 \text{ df}$ $p = 0.38$
<i>Difference between 2010 match days and 2009 control</i>				
B: Proportional change in rate	5.2% $t = 1.49, 29 \text{ df}$ $p = 0.153$	5.1% ^a $Z = -1.31, n = 30$ $p = 0.192$	35.4% $t = 5.87, 29 \text{ df}$ $p < 0.001$	33.9% $t = 6.16, 29 \text{ df}$ $p < 0.001$
<i>Difference between 2010 match days and 2010 control</i>				
C: Proportional change in rate	-1.91% $t = -0.26, 32 \text{ df}$ $p = 0.797$	0.1% $t = 0.57, 32 \text{ df}$ $p = 0.595$	27.7% $t = 7.07, 32 \text{ df}$ $p < 0.001$	31.5% $t = 6.39, 32 \text{ df}$ $p < 0.001$

^a Wilcoxon signed-rank test due to non-normality of the data